Section 10: Data Structures, Vectors and Stacks Extension

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Standard Template Library (STL)

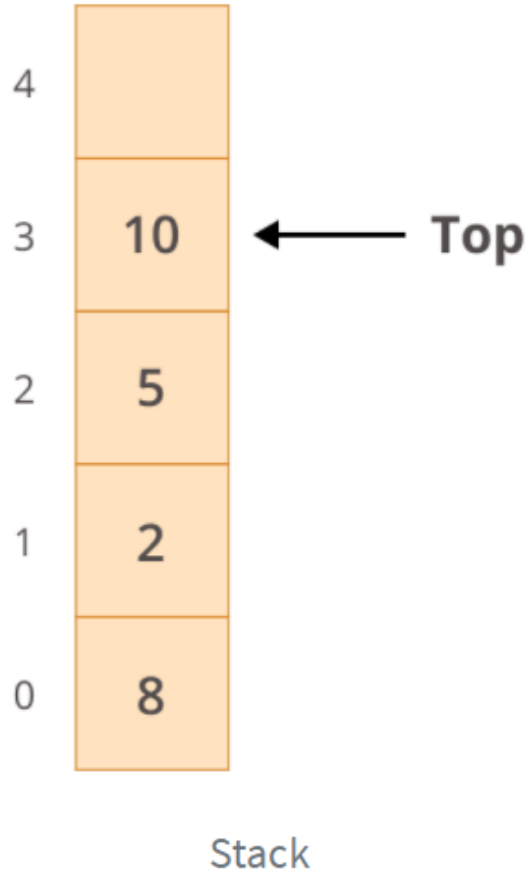
- Standard Template Library (STL) is a collection of template classes that provide data structures such as arrays, vectors, queues, etc.

- STL mainly consists of three components:
  - Containers
    - Containers manage the collection of elements, e.g. array, vector, stack …
  - Algorithms
    - Algorithms are the methods or functions that act on containers, e.g. search, sort, modify…
  - Iterators
    - Iterators are the constructs that are used to traverse through the container objects.

https://en.cppreference.com/w/cpp

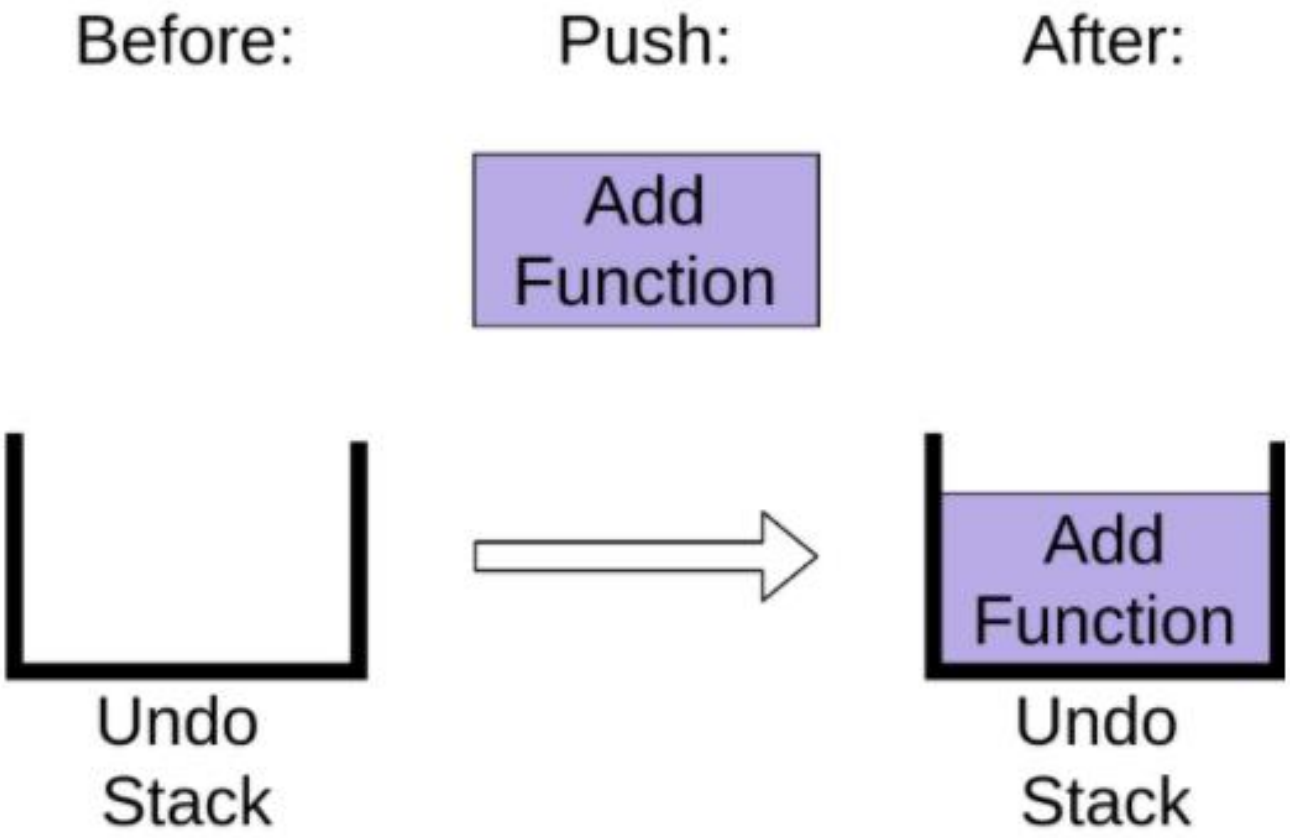
Stack

- a stack: **Last-In/Fist-Out (LIFO)** approach
  - the last element inserted (at the top) will be the first to be retrieved.



Undo feature in an editor:

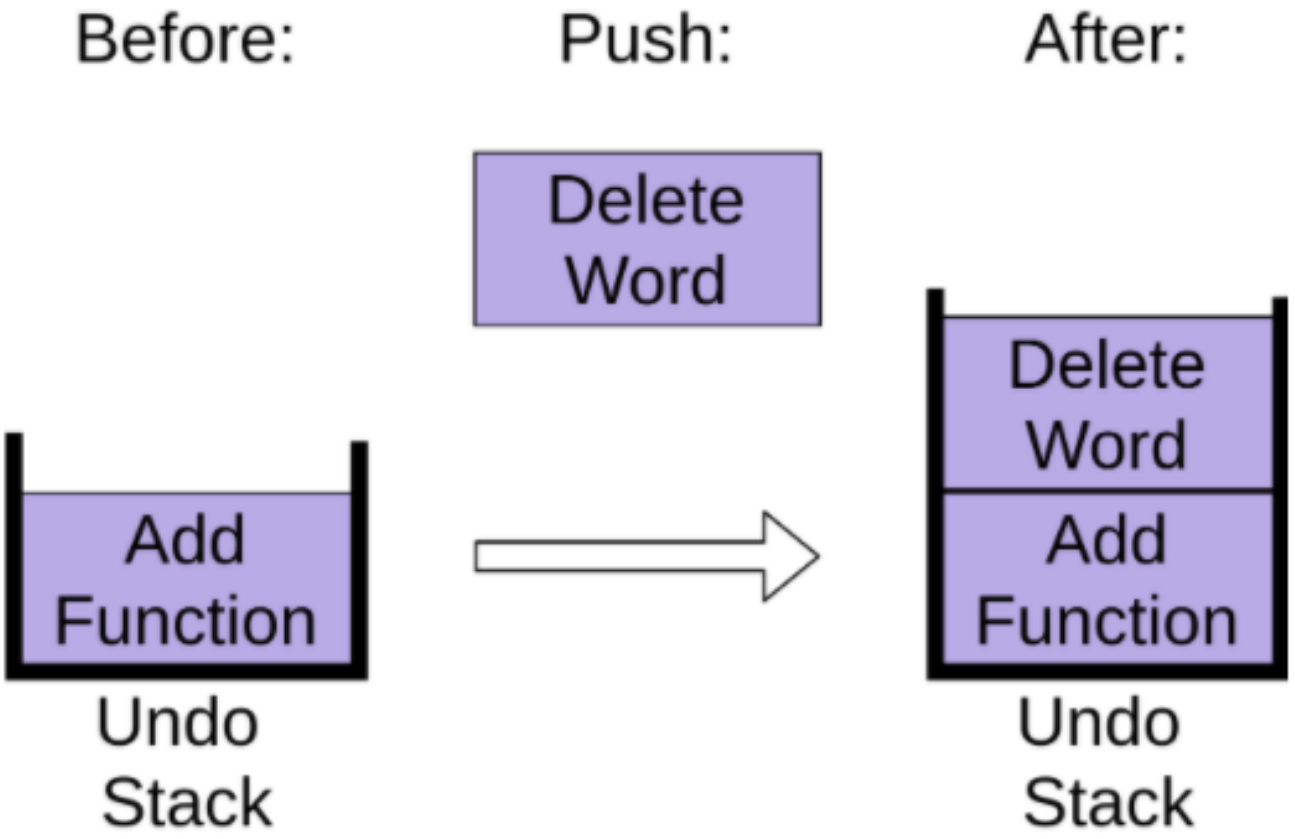
Add “Add Function” to Undo Stack



https://realpython.com/how-to-implement-python-stack/

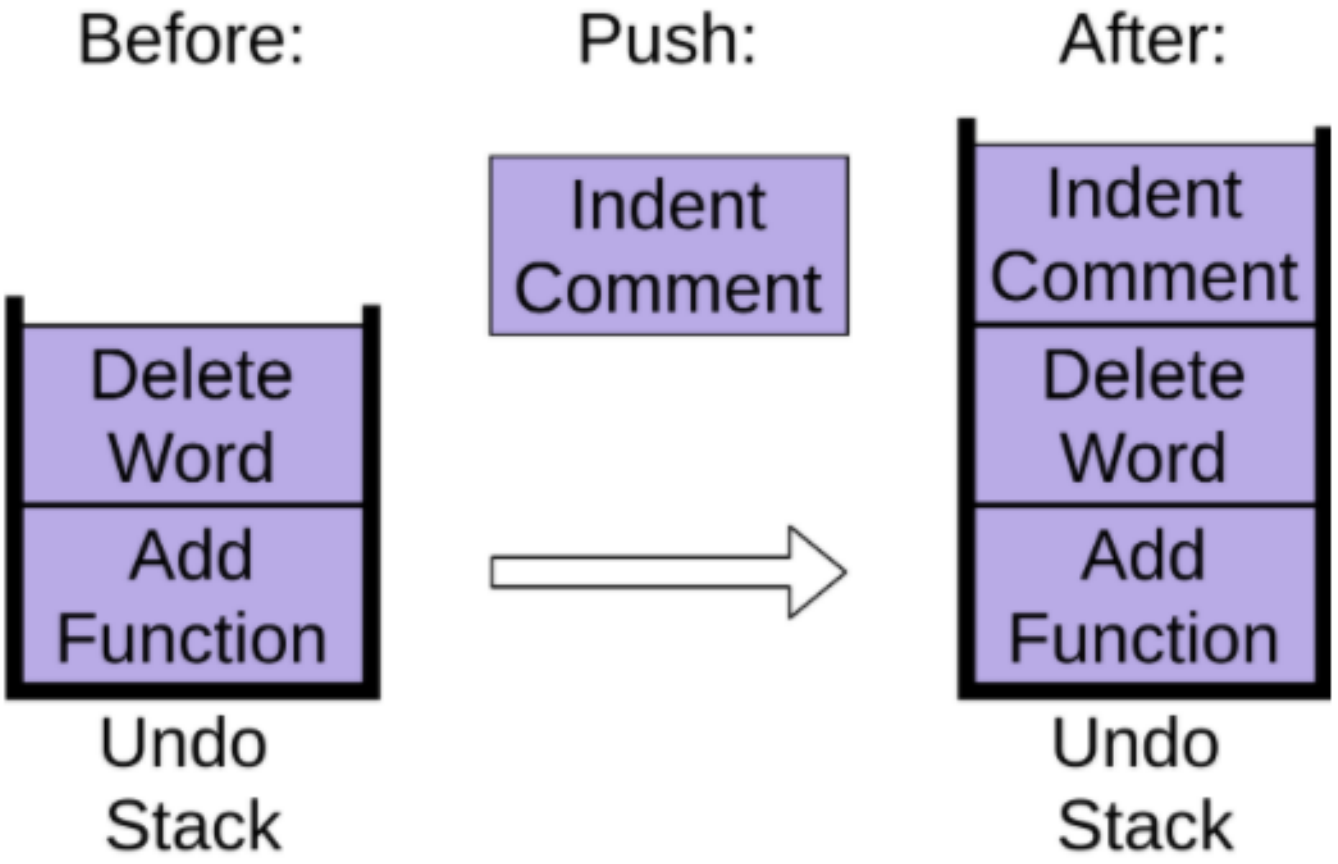
Undo feature in an editor:

Add “Delete Word” to Undo Stack



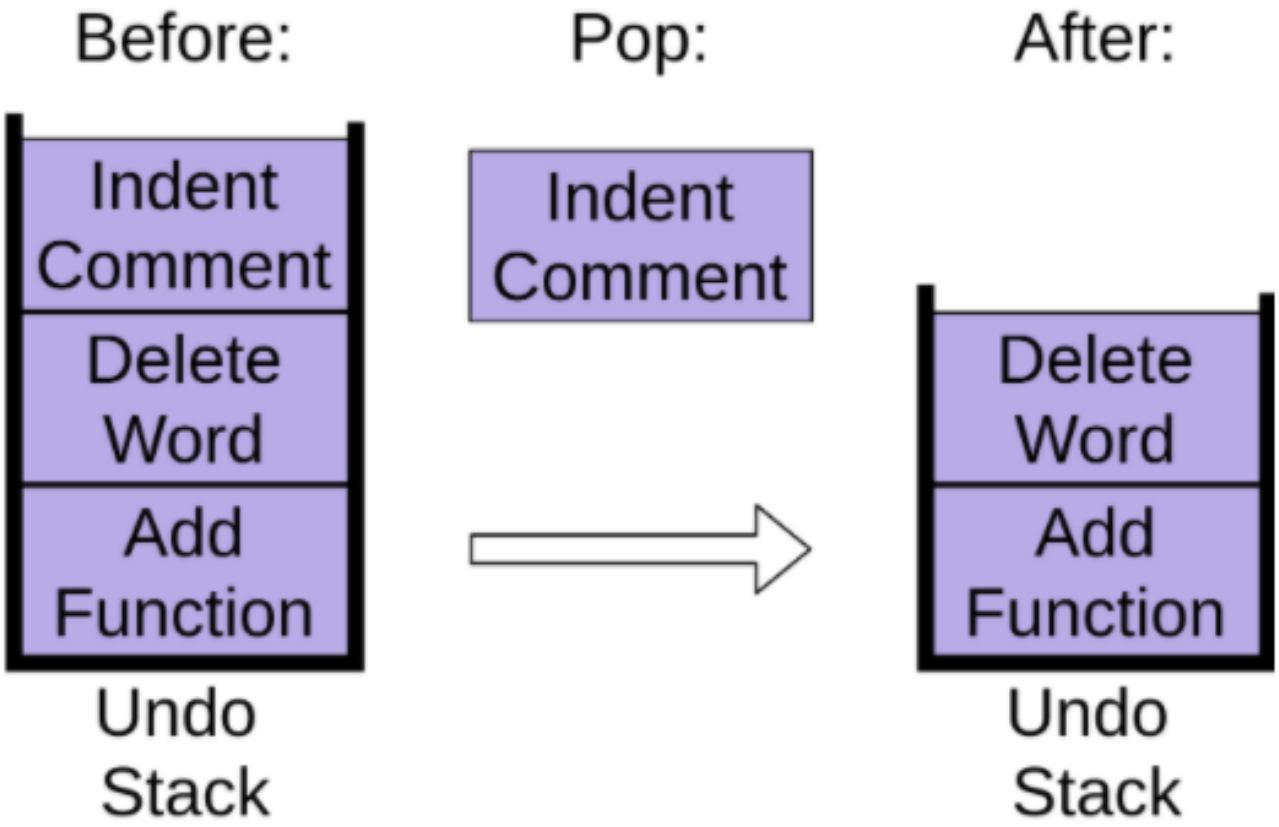
Undo feature in an editor:

Add “Indent Comment” to Undo Stack



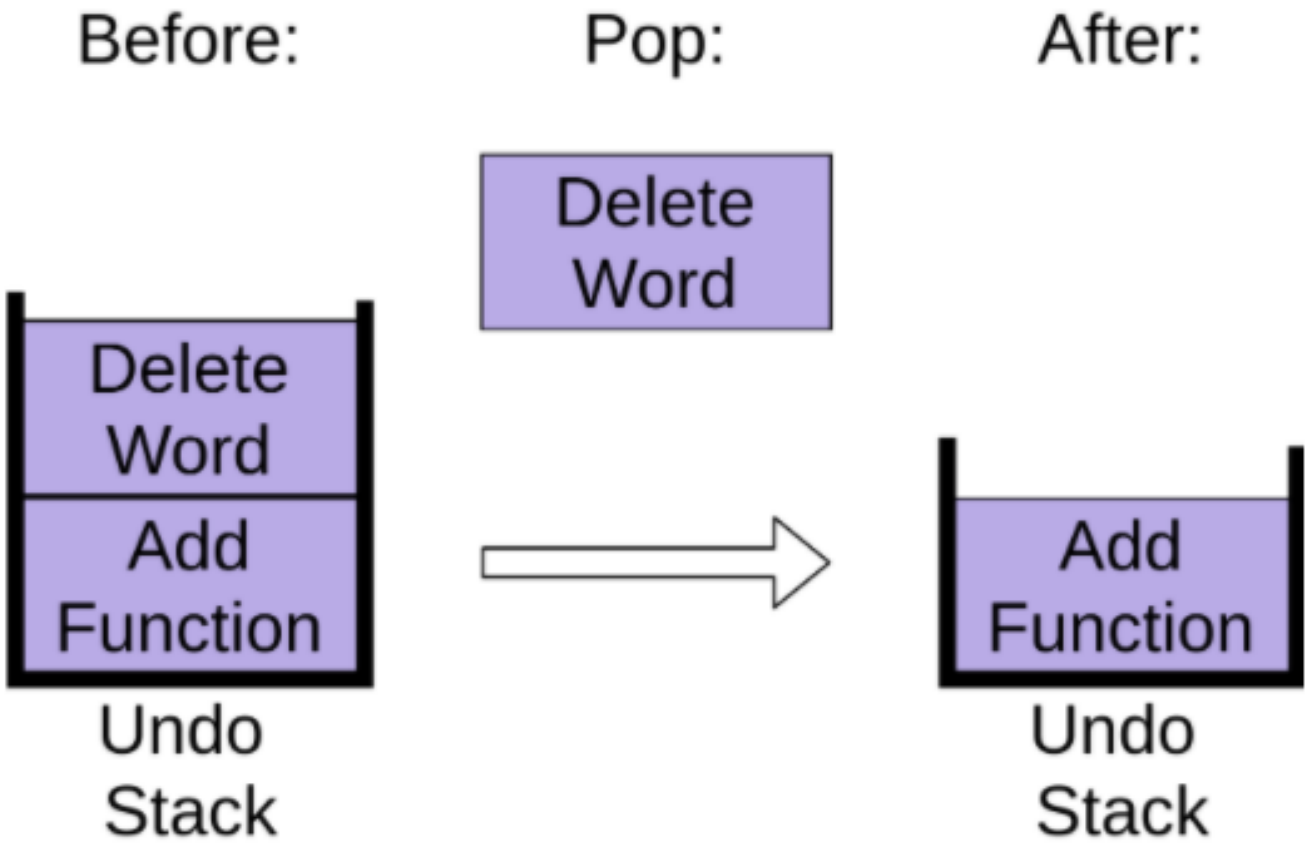
Undo feature in an editor:

Undo “Indent Comment”



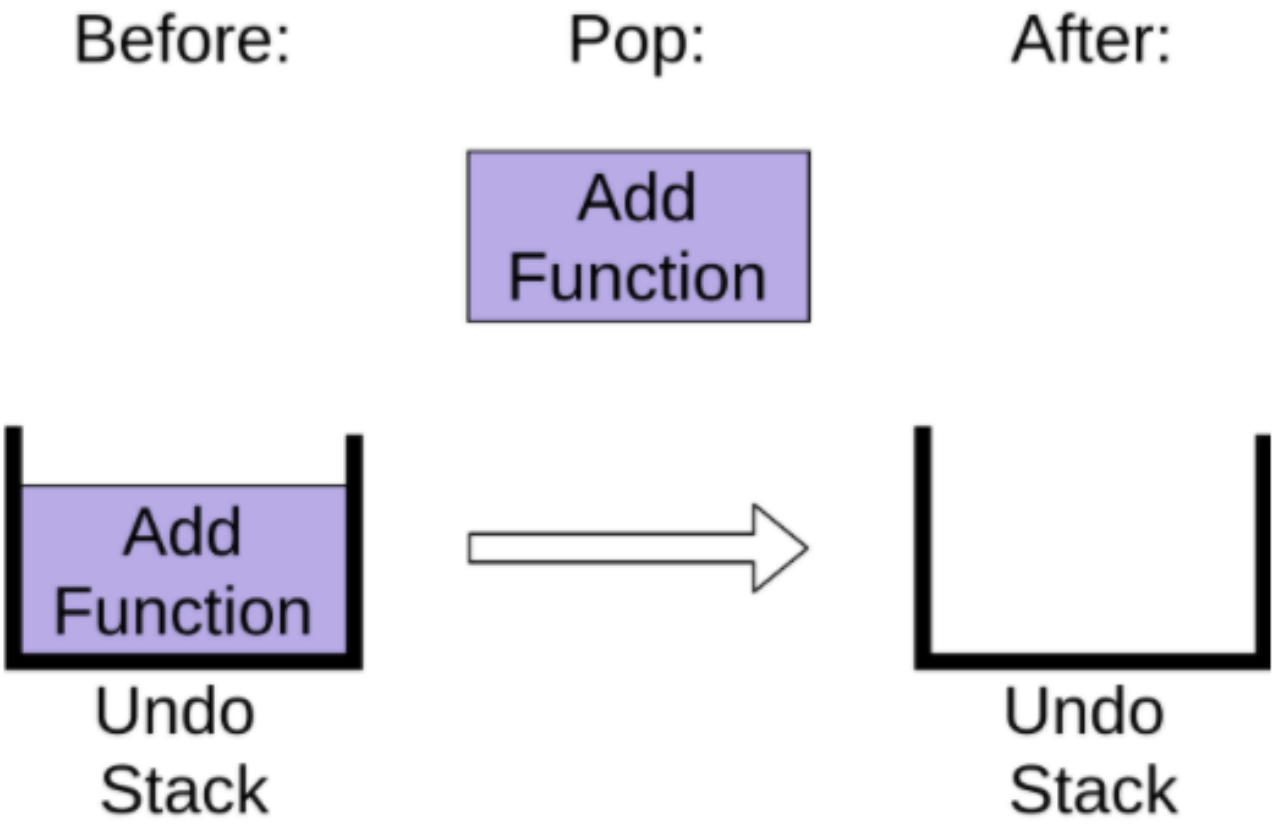
Undo feature in an editor:

Undo “Delete Word”



Undo feature in an editor:

Undo “Add Function”

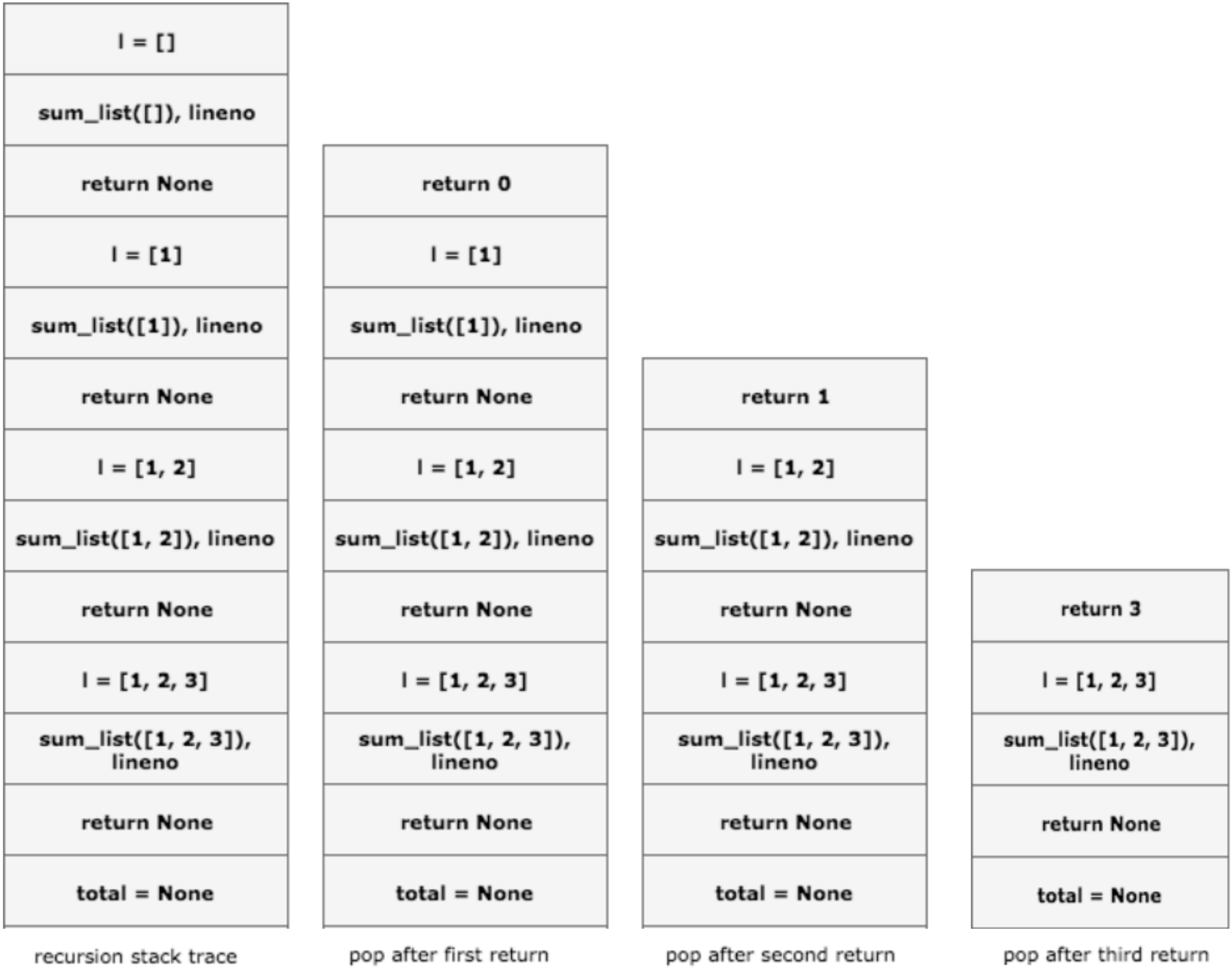


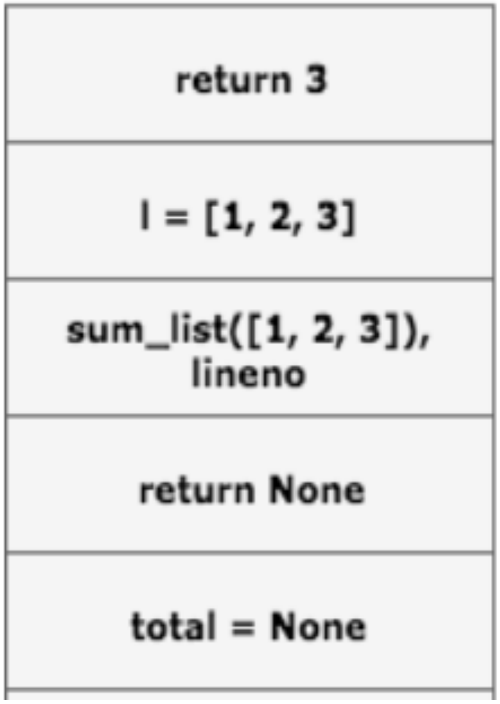
Recursion works on run-time stack

```
int sum_list(int l[], int n)
{
    if (n <= 0)
        return 0;
    return (l[n-1] + sum_list(l, n - 1));
}

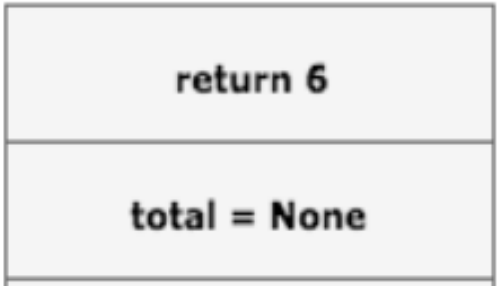
int b[3]={1,2,3};
int total=sum_list(b,3);
```

https://www.codementor.io/@ferozkhan27/in-python-how-recursion-works-on-run-time-ymd54rw4h

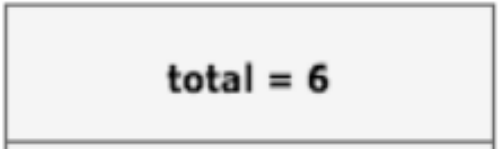

l = []
sum_list([]), lineno
return None
l = [1]
sum_list([1]), lineno
return None
l = [1, 2]
sum_list([1, 2]), lineno
return None
l = [1, 2, 3]
sum_list([1, 2, 3]), lineno
return None
total = None
recursion stack trace
return 0
l = [1]
sum_list([1]), lineno
return None
l = [1, 2]
sum_list([1, 2]), lineno
return None
l = [1, 2, 3]
sum_list([1, 2, 3]), lineno
return None
total = None
pop after first return
return 1
l = [1, 2]
sum_list([1, 2]), lineno
return None
l = [1, 2, 3]
sum_list([1, 2, 3]), lineno
return None
total = None
pop after second return
return 3
l = [1, 2, 3]
sum_list([1, 2, 3]), lineno
return None
total = None
pop after third return



pop after third return



pop after fourth return



pop after fifth return